

Yacouba OUEDRAOGO

AI & Climate Data Scientist –
Machine Learning for Climate
Systems

CONTACT

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[GitHub](#) || [LinkedIn](#)

LANGUAGES

French: Native Speaker
English: Good

COMPUTER SKILLS

- Python, R, C#, SQL
- ArcMap, QGIS
- Google Earth Engine
- Tableau software
- Power BI Desktop
- Latex
- CDO
- ODK, Kobo

Awards

- *Alumni Excellence Award – AIMS Senegal (2025)*
- *Small innovation grants program (USAID/NASA)*
- *MS4CR Scholarship*
- *Master Scholarship of AIMS*
- *WANECAM Scholarship*

Center of Interest

- Data coding and analysis
- Early Warning

Professional Summary

Climate Data Scientist with 6 years of experience designing and deploying machine learning solutions for climate and environmental data systems in Africa. Specialized in AI-enabled data ingestion, quality control, gap-filling, and climate analytics using large-scale geospatial and time-series datasets (CHIRPS, ERA5, Copernicus, ACLED). Proven experience building reproducible AI/ML pipelines, integrating models into operational platforms, and supporting early-warning and decision-support systems. Strong background in R and Python, open-source MLOps practices, and capacity building through technical documentation, SOPs, and staff training.

Key Achievements

- **CRAFT automation (100%)**: Fully automated climate data ingestion into CRAFT, enabling scheduled and reproducible end-to-end runs.
- **>90% runtime reduction**: Replaced ArcGIS geoprocessing with open-source R/Python pipelines (terra, sf, raster), cutting compute time by over 90%.
- **Open-sourced CRAFT**: Removed proprietary dependencies and delivered a portable, reproducible open-source stack with lower total cost of ownership.
- **Multi-source pipelines**: Engineered R/Python workflows integrating 30+ years of CHIRPS and ERA5 climate data, ACLED conflict data, and Cadre Harmonisé food security indicators across 8 Sahel countries.
- **Composite indices & mapping**: Built PCA-based conflict-climate indices and 20×20 km risk grids; produced vulnerability maps used by partners (WFP, UNHCR, CIAT).
- **Crop yield prediction (ML)**: Built and validated machine-learning models (Random Forest, GAM) using climate and EO features (rainfall anomalies, SPEI-3, temperature extremes, NDVI/EVI) to forecast district-level crop yields.

EDUCATION

Sept 2024	Kalu Institute
Mars 2026	Master’s in international Cooperation and Humanitarian Aid
Nov 2020	AIMS - SENEGAL, (center of excellence in Africa)
Jul 2022	Master’s in mathematical sciences: Big Data/Machine Learning
Jun 2016	Université Nazi Boni
Sep 2019	Master of Applied Mathematics: Modelling and Scientific Computing

PROFESSIONAL EXPERIENCES

RESEARCH ASSOCIATE - GIS AND CLIMATE DATA SCIENCE at Alliance Bioversity & CIAT (Senegal) || Feb 2024-Present

- Designed and operationalized a geospatial early-warning product combining climate anomalies and conflict intensity, enabling WFP to trigger anticipatory actions in 32 high-risk districts across 8 Sahelian countries.
- Served as technical focal point for UNHCR, coordinating climate–conflict analytics for displacement risk, including a 20 × 20 km gridded methodology integrating socio-health, security, and service-access indicators with climate stress metrics (SPEI-3).
- Applied machine-learning and geospatial methods (Random Forest, Gradient Boosting, GAMs, cPCA, K-means) on satellite, sensor, and field data using reproducible Python pipelines.
- Conducted localized climate-risk assessments for the Climate Security Observatory, including scenario-based analyses in fragile and conflict-affected settings.
- Co-authored technical reports and geospatial briefs and contributed to proposals on climate security and food systems

Risk Data Specialist (Consultant) – World Bank | 2025 – Present

- Supported climate and conflict risk analytics for decision support and early warning.
- Conducted data readiness assessments and ensured compliance with data governance and privacy standards.

RESEARCH FELLOW- Spatial Analyst at University of Florida (USA-Gainesville)

|| Oct 2022 - Feb 2024

- Automated ingestion, harmonization, and QA of gridded climate datasets (NASA POWER, Copernicus, CHIRPS) into the CCAFS Regional Agricultural Forecasting Toolbox (CRAFT), reducing manual processing time by ~80% and enabling scalable workflows.
- Built and validated DSSAT-based crop production hindcasts at district level for key crops in Ghana, assessing predictive performance under varying climate and soil conditions.
- Developed modular workflows integrating EO-based crop and crop-type masks into CRAFT to enable finer spatial analyses across West Africa.
- Conducted sensitivity analyses to quantify the impact of EO data uncertainty on model outputs, improving forecast robustness and interpretability

VISITING RESEARCH AT NASA (USA) | | Jan 2024

- Prototyped near-real-time workflows generating dynamic crop production forecasts using Earth observation data streams.

CONSULTANT, DATA QUALITY & METHODOLOGY at IDEMS International

| | Jan 2023 -Apr 2023

- Built automated QA/QC pipelines for ODK/Kobo data (range and consistency checks, skip logic, GPS validation, deduplication), reducing data-cleaning time and improving first-pass acceptance.
- Developed R-based mapping and batch-export pipelines, shortening turnaround from raw surveys to decision-ready maps.

GIS DATA SCIENTIST(Intern) at African Centre of Meteorological Applications for Development (ACMAD) (Niger) | | Apr 2022 - Sep 2022

- Automated agro-climatic bulletins (onset, drought, dry spells) for the African region, with standardized and routine updates.
- Delivered climate–health early-warning analytics (malaria, meningitis) using rainfall, temperature, humidity, and epidemiological indicators.
- Produced regional geodatabases of indicators for environmental risk assessment.

CONSULTANT- GIS ANALYST | CLIMATE RISK at Red Cross Red Crescent Climate Centre | | Mar 2022 - May 2022

- Analyzed public and UNICEF-provided flood impact data to identify high-risk areas in Kaduna State, Nigeria.
- Produced flood risk maps supporting evidence-based planning and anticipatory action for UNICEF’s humanitarian response.
- Synthesized findings into a concise technical report and stakeholder-oriented presentations.

JUNIOR RESEARCHER | FORECAST-BASED FINANCING (FBF) & DISASTER RESILIENCE at Red Cross Red Crescent Climate Centre (Senegal) | | Oct 2021 - Feb 2022

- Developed AI-based crop yield prediction models using climate variables to support anticipatory and resilience-oriented planning.
- Supported national climate risk assessments (hazards, exposure, vulnerability) using multi-source data and expert inputs.
- Evaluated forecast skill (GloFAS/ECMWF, CHIRPS/ERA5) to inform FbF activation thresholds, documenting lead times and uncertainty.

- Contributed to improvements of Early Action Protocols (EAPs), aligning forecast triggers with operational decision timelines.

BIostatistician | Health Data Analyst at INSTech (Institut des Sciences et Techniques, Burkina Faso) || Jan 2019 - Oct 2020

- Analyzed clinical data on malaria treatment effectiveness, supporting public health decision-making.
- Led data quality assurance and survey data analysis, improving research reliability

SKILLS:

AI & Machine Learning

- ML models for climate data (Random Forest, GAM, PCA)
- Anomaly detection, gap-filling, bias correction
- Climate indices and trend analysis (SPEI-3, rainfall, drought)

Data Pipelines & Engineering

- Automated ingestion, QC, and validation pipelines
- Large-scale geospatial and time-series processing
- Reproducible open-source workflows (R, Python)

Climate & EO Data

- CHIRPS, ERA5, Copernicus, NASA POWER
- EO indicators (NDVI, EVI, temperature extremes)
- Conflict and food security data (ACLED, IPC)

CERTIFICATIONS

- Project Management from Kalu Institute
- NASA's Applied Remote Sensing Training (ARSET): Fundamentals of Machine Learning for Earth Science
- Humanitarian Leadership Academy, Humentum, Catholic Relief Services (CRS): MEAL DPro: Monitoring, Evaluation, Accountability and Learning for Development Professionals
- YONSEI University: Spatial Data Science and Applications